

Sara Ramaiah

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SUMMARY

Bioinformatics research scientist with over 2 years full-time equivalent experience. Specializes in bridging the gap between bioinformatics and software engineering, with broad proficiencies in biological wet lab research in pathology/immunology.

WORK EXPERIENCE

CASE WESTERN RESERVE UNIVERSITY, DEPARTMENT OF PATHOLOGY
Research Assistant

Cleveland, OH
Sept. 2022-May 2025

- Led the lab's computational biology development, including configuring AWS resources, packaging stable implementations of core software, and developing next-generation sequencing analysis pipelines.
- Spearheaded analysis of single cell RNA sequencing in R, and 16S gut microbial sequencing in Python with Nextflow.
- Designed exploratory analysis of Alzheimer's Disease patient sample data via applications of graph data science to the Reactome database and scikit-learn predictive analysis methods.
- Managed multiple undergraduate students in a collaborative project, fostering good lab practices and communication.

CASE WESTERN RESERVE UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE
Teaching Assistant, CSDS 310: Algorithms

Cleveland, OH
Aug. 2023 - Dec. 2023

- Taught a class of 20 students weekly, focusing on translating theory into actionable problem-solving skills.
- Delivered detailed explanations in office hours and detailed feedback on assignments for students needing support, reframing and communicating key concepts to a diverse array of student needs.

UNIVERSITY HOSPITALS, CLEVELAND MEDICAL CENTER
Patient Care Nursing Assistant

Cleveland, OH
Jan. 2024 - July 2024

- Cared for up to 15 patients at once in high acuity medical-surgical inpatient care.
- Used Epic Electronic Health Record system to securely track and record patient medical information.
- Excelled at communicating disease and medical treatment biology to non-technically oriented patients.

CASE WESTERN RESERVE UNIVERSITY, DEPARTMENT OF BIOCHEMISTRY
Summer Undergraduate Research Student

Cleveland, OH
May 2022 - Aug. 2022

- Studied RNA-binding proteins in naive and mature T cells via cell culture with Jurkat cell line.
- Designed DNA plasmids and restriction enzyme screening protocols for recombinant protein synthesis with *E. coli*.

EDUCATION

CASE WESTERN RESERVE UNIVERSITY
Bachelor of Science, Biochemistry

Cleveland, OH
2020 - 2024

- Minors in Computer Science and Mathematics
- Cumulative GPA: 3.97

SKILLS

Programming languages: Python, R, Java, Nextflow, SQL, Cypher, Nix

Software: Linux, Docker, Git, AWS, NixOS, scikit-learn, Neo4j GDS library, Sequera, GraphPad Prism, Microsoft Office

Bioinformatics: High Performance Computing, multi-omics analysis, genome sequencing, metagenomics, nf-core pipelines

Biochemistry: Western blotting, Nucleic acid extraction, DNA sequencing, protein purification, cell culture

PROJECTS

Declarative Environments for Bioinformatics

- Developed strongly reproducible bioinformatics environments via the Nix package manager.
- Published Docker image and compose file for basic bioinformatics packages, coupled with Jupyter notebooks. Partnered with a student organization to run an educational session with undergraduates using this tutorial resource.
- Used these environments for scRNA-seq, 16S sequencing analysis, Reactome database hosting and analysis, and more.

IL17-dependent inflammation in C9orf72-ALS/FTD

- Led bioinformatics branch of study exploring the relationship between C9orf72 and IL-17A in a mouse model.
- Ran experiments such as single-cell sequencing, Western blots, behavioral testing, and tissue histology.
- Piloted scRNA-seq analysis protocol for the lab via this experiment's dataset, including my expertise in declarative environments and package management for optimal reproducibility in future experiments.